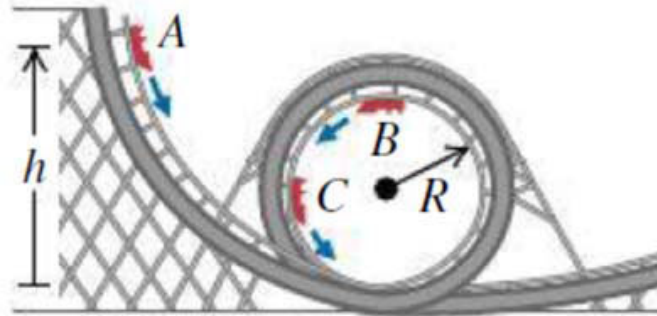
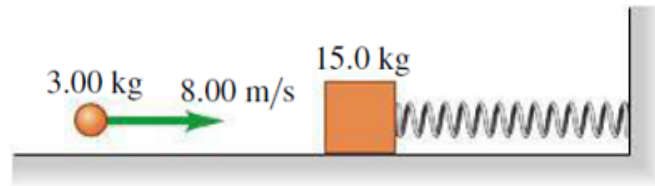


1. A car in an amusement park ride rolls without friction around a track.



The car starts from rest at point A at a height h above the bottom of the loop. Treat the car as a particle.

- (i) What is the minimum value of h in terms of R such that the car moves around the loop without falling off at the top (point B)? $\left(\frac{5}{2}R\right)$
 - (ii) If $h = 4.50R$ and $R = 16.0$ m, compute the speed, radial acceleration, and tangential acceleration of the passengers when the car is at point C , which is at the end of a horizontal diameter. Show these acceleration components in a diagram, approximately to scale. $(\sqrt{7gR}, -, -)$
2. A wooden block with mass 1.50 kg is placed against a compressed spring at the bottom of an incline of slope 30.0° (point A). When the spring is released, it projects the block up the incline. At point B , a distance of 6.00 m up the incline from A , the block is moving up the incline at 7.00 m/s and is no longer in contact with the spring. The coefficient of kinetic friction between the block and the incline is $\mu_k = 0.50$. The mass of the spring is negligible. Calculate the amount of potential energy that was initially stored in the spring. (119.04 J)
3. A 15.0 -kg block is attached to a very light horizontal spring of force constant 575.0 N/m and is resting on a frictionless horizontal table. Suddenly it is struck by a 3.00 -kg stone travelling horizontally at 8.00 m/s to the right, whereupon the stone rebounds at 2.00 m/s horizontally to the left.



Find the maximum distance that the block will compress the spring after the collision. (0.32 m)

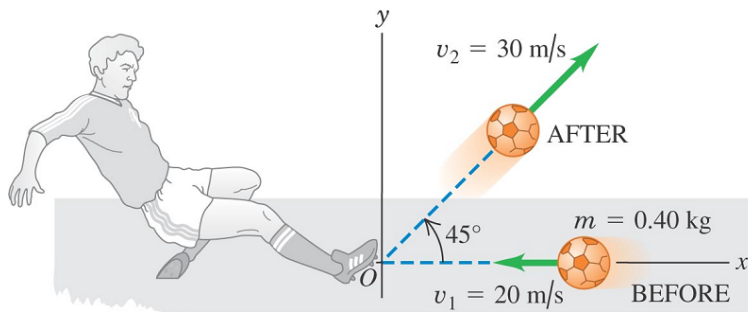
4. A 22.00-kg lead sphere is hanging from a hook by a thin wire 4.00 m long and is free to swing in a complete circle. Suddenly it is struck horizontally by a 3.00-kg steel dart that embeds itself in the lead sphere. What must be the minimum initial speed of the dart so that the combination makes a complete circular loop after the collision? (116.67 m/s)
5. A 20.0-kg projectile is fired at an angle of 60.0° above the horizontal with a speed of 80.0 m/s. At the highest point of its trajectory, the projectile explodes into two fragments with equal mass, one of which falls vertically with zero initial speed. Ignore air resistance.
 - (i) How far from the point of firing does the other fragment strike if the terrain is level? (848.35 m)
 - (ii) How much energy is released during the explosion? (16000 J)
6. Consider three objects of masses m_1 , m_2 and m_3 whose positions are on a straight horizontal line as shown. The surface is frictionless and all collisions are elastic. The objects m_2 and m_3 are at rest initially, and the object m_1 is moving towards object m_2 with a speed of v_0 . The 3 masses have different magnitudes.



- (i) What is the velocity (in terms of m_1 , m_2 , m_3 and v_0) of m_3 after being hit by m_2 ?
- (ii) If the magnitude of m_2 can be changed, what will be the value of m_2 (in terms of m_1 and m_3) for the maximum energy transfer from m_1 to m_3 ? ($m_2 = \sqrt{m_1 m_3}$)

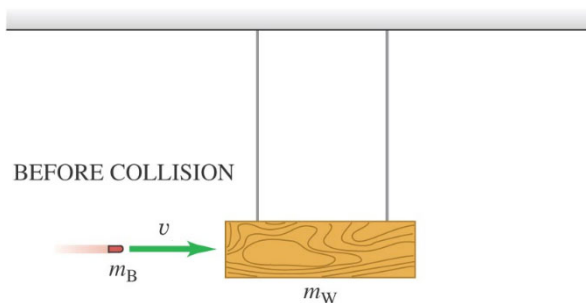
Note: In this question m_1 , m_2 , m_3 index the objects, and also represent the respective mass of the objects.

7. A soccer ball has a mass of $m = 0.400$ kg. Initially it is moving to the left at $v_1 = 20.0$ m/s, but then it is kicked. After that kick it is moving 45° upward and to the right with speed $v_2 = 30.0$ m/s.



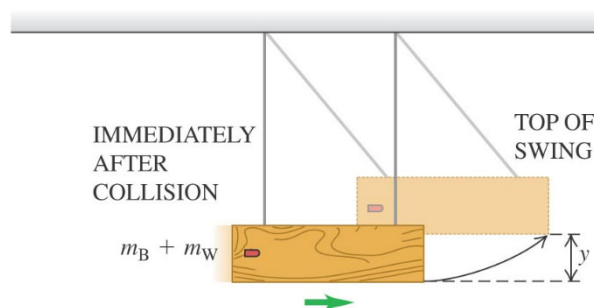
Find the impulse of the net force and the average net force along the x -direction, and along the y -direction, assuming a collision time $\Delta t = 0.0100$ s. (16.49 kg m/s, 8.49 kg m/s, 1649 N, 849 N)

8. The figure shows a ballistic pendulum, a simple system for measuring the speed of a bullet.



A bullet of mass m_B makes a completely inelastic collision with a block of wood of mass m_W , which is suspended like a pendulum. After the impact, the block swings up to a maximum height y . In terms of y , m_B , and m_W , what is the initial speed v of the bullet?

$$\left(\left(1 + \frac{m_W}{m_B} \right) \sqrt{2gy} \right)$$



9. If you have ever stepped on a small boat, you have probably noticed that the boat moves away from the dock as you step towards the dock or out of the boat. A lovely dog running on a long lightweight raft presents a similar situation as yours. At first, the raft and the lovely dog are at rest with respect to the water as shown in the Figure (a).

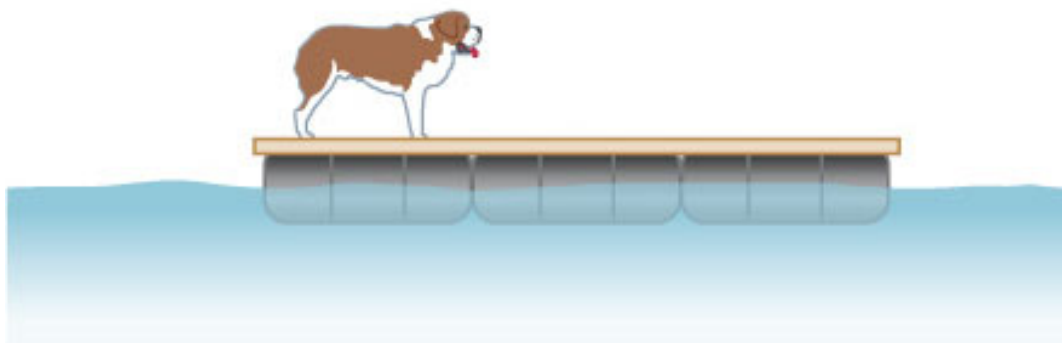


Figure (a)

The lovely dog then runs on top of the raft at $\vec{v}_{\text{dog}} = v_{\text{dog}} \hat{i}$ with respect to the water (Figure (b)). The lovely dog has twice the mass of the raft. What is the velocity of the raft after the lovely dog began running?

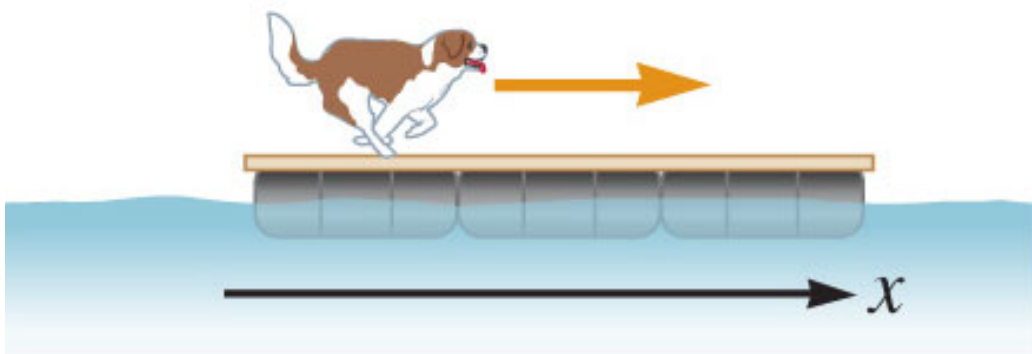
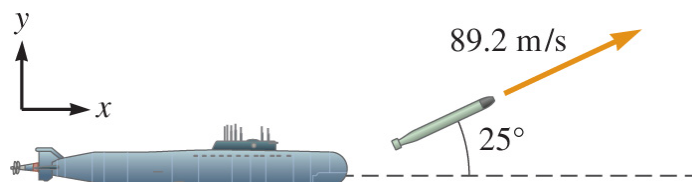
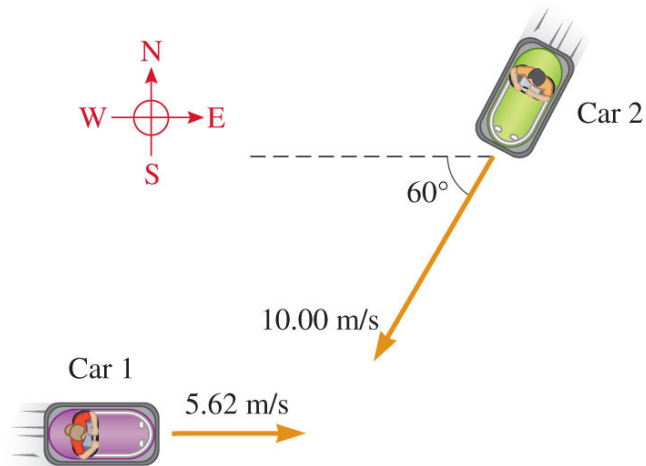


Figure (b)

10. A submarine with a mass of 6.26×10^6 kg contains a torpedo with a mass of 354 kg. The submarine fires the torpedo at an angle of 25° with respect to the horizontal as shown in the figure.



- (i) If the submarine and the torpedo were initially at rest and the torpedo left the submarine with a speed of 89.2 m/s, what is the recoil speed of the submarine?
(5.042 x 10⁻³ m/s)
- (ii) What is the direction of recoil of the submarine?
11. Two bumper cars at the country fair are sliding toward one another as shown in the figure. Initially, bumper car 1 is travelling to the east at 5.62 m/s, and bumper car 2 is travelling 60.0° south of west at 10.00 m/s. After they collide, bumper car 1 is observed to be travelling to the west with a speed of 3.14 m/s. Friction is negligible between the cars and the ground.

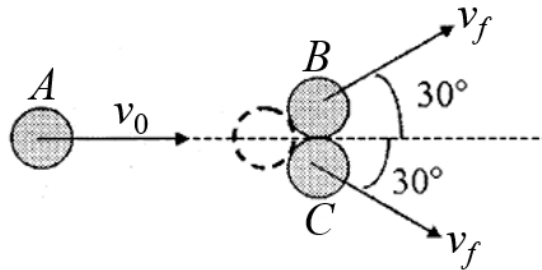


- (i) If the masses of bumper cars 1 and 2 are 596 kg and 625 kg respectively, what is the velocity of bumper car 2 immediately after the collision?

$$\left((3.35\hat{i} - 8.86\hat{j}) \text{ m/s} \right)$$

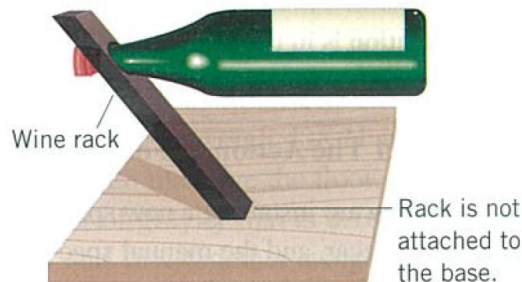
- (ii) What is the kinetic energy lost in the collision? (10780.83 J)

12. Ball A , sliding with an initial speed $v_0 = 10.0 \text{ m/s}$ on a frictionless surface, collides elastically with balls B and C as shown in the figure. All the balls are identical. If balls B and C move off with the same speed v_f at 30° as indicated below where $v_f \neq 0$,



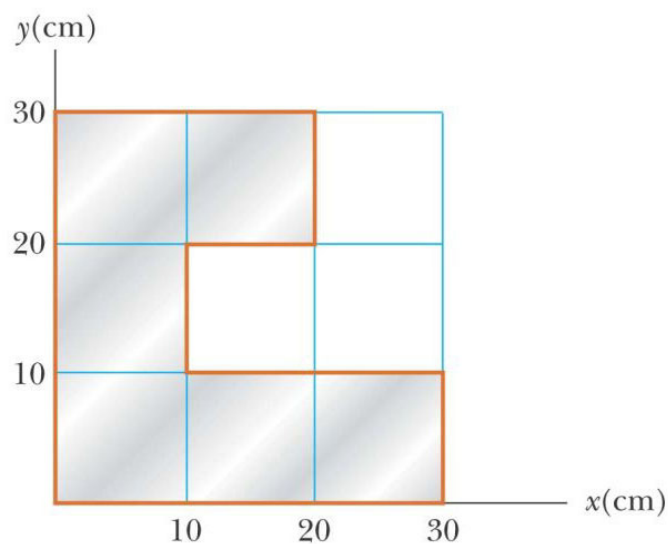
- (i) prove that the speed of ball A after the collision cannot be 0,
 (ii) calculate the numerical value of v_f , and (6.93 m/s)
 (iii) calculate the magnitude and direction of velocity for ball A after the collision. (2 m/s)

13. Many of you may have seen this wine rack for a single bottle of wine that seems to defy our common sense as it balances on any flat surface.



The wine track, i.e., the wooden stick, is actually not inserted into the base. Treat the stick and wine bottle as a single rigid body and draw the forces that keep it in equilibrium. In particular, where must the center of mass of the rigid body be located? Give your reasoning.

14. A uniform piece of sheet steel is shaped as shown in the figure. Compute the x and y coordinates of the center of mass of the piece. (11.67 cm, 13.33 cm)



Formula List

Logarithm:

$$\log a + \log b = \log(a \times b)$$

$$\log a - \log b = \log\left(\frac{a}{b}\right)$$

$$(\log x) > 0, \text{ if } x > 1.$$

$$(\log 1) = 0.$$

$$(\log x) < 0, \text{ if } 0 < x < 1.$$

$$(\log x) \text{ is undefined if } x < 0.$$

Trigonometry:

$$\sin = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan = \frac{\text{opposite}}{\text{adjacent}} = \frac{\sin}{\cos}$$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

$$\cos 2\theta = 1 - 2 \sin^2 \theta$$

$$\cos 2\theta = 2 \cos^2 \theta - 1$$

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

Differentiation:

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}(e^x) = e^x$$

$$\frac{d}{dx}(e^{f(x)}) = e^{f(x)} \times \frac{d}{dx}(f(x))$$

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(f(x) \times g(x)) = f(x) \times \left(\frac{d}{dx}g(x)\right) + \left(\frac{d}{dx}f(x)\right) \times g(x)$$

Integration:

$$\int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + c$$

Vector:

$$\vec{A} + \vec{B} = \vec{B} + \vec{A}$$

$$\vec{A} \bullet \vec{B} = |\vec{A}| \times |\vec{B}| \times \cos \theta$$

Linear kinematic equations:

$$v_f = v_i + at$$

$$\Delta x = v_i t + \frac{1}{2} at^2$$

$$\Delta x = \frac{(v_i + v_f)}{2} \times t$$

$$v_f^2 = v_i^2 + 2a\Delta x$$

Simple harmonic motion:

$$x(t) = r_0 \cos(\theta_0 + \omega_0 t)$$

$$v_x(t) = -r_0 \omega_0 \sin(\theta_0 + \omega_0 t)$$

$$a_x(t) = -r_0 \omega_0^2 \cos(\theta_0 + \omega_0 t) \\ = -\omega_0^2 x(t)$$

$$\omega^2 = \frac{k}{m}$$

$$T = 2\pi \sqrt{\frac{m}{k}}$$

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

$$v_{\max} = \sqrt{\frac{k}{m}} A$$

$$a_{\max} = \frac{k}{m} A$$

Retarding force:

$$R = -bv$$

Projectile motion:

$$y = -\frac{g}{2v_{0x}^2}(x - x_0)^2 + \frac{v_{0y}}{v_{0x}}(x - x_0) + y_0, \text{ or}$$

$$y = x \tan \theta - \frac{gx^2}{2v_0^2}(1 + \tan^2 \theta)$$

Uniform circular motion:

$$\theta = \frac{\text{arc length}}{\text{radius}} = \frac{s}{r}$$

$$\omega = \frac{d\theta}{dt} = \frac{2\pi}{T}$$

$$v = r\omega$$

$$a_c = r\omega^2 = \frac{v^2}{r}$$

Non-uniform circular motion:

$$a_t = \frac{dv}{dt}$$

$$a = \sqrt{a_c^2 + a_t^2}$$

Relative velocity:

$$\vec{V}_{P/A} = \vec{V}_{P/B} + \vec{V}_{B/A}$$

Newton's second law:

$$\sum \vec{F} = m\vec{a}$$

Newton's third law:

$$\vec{F}_{12} = -\vec{F}_{21}, \text{ and the two forces} \\ \text{act on the opposite body.}$$

Frictional force:

$$f_s \leq \mu_s n, \text{ and impending motion} \\ \text{occurs when } f_s = \mu_s n.$$

$$f_k = \mu_k n$$

Translational kinetic energy:

$$KE_{\text{translational}} = \frac{1}{2}mv^2$$

Gravitational potential energy:

$$\Delta PE = mg(y_2 - y_1) \\ = mgh$$

Spring

$$F_{\text{spring}} = -kx$$

$$\text{Elastic } PE_{\text{spring}} = \frac{1}{2}kx^2$$

Work-energy theorem:

$$W = \Delta KE = KE_f - KE_i$$

Work done (constant force):

$$W = \vec{F} \bullet \Delta \vec{r} = F \Delta r \cos \theta$$

Work done (varying force):

$$W = \int_{x_i}^{x_f} \vec{F} \bullet d\vec{r}$$

Power:

$$P = \frac{dW}{dt} = F \times v$$

Linear momentum:

(also valid for y and z)

$$p_x = mv_x$$

$$F_x = \frac{dp_x}{dt}$$

$$F_{\text{average}, x} = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} F_x(t) dt$$

Conservation of linear momentum:

$$\vec{p}_{1,i} + \vec{p}_{2,i} = \vec{p}_{1,f} + \vec{p}_{2,f}$$

Elastic collision:

$$v_{A1} - v_{B1} = -(v_{A2} - v_{B2})$$

$$v_{A2} = \left(\frac{m_A - m_B}{m_A + m_B} \right) v_{A1} + \left(\frac{2m_B}{m_A + m_B} \right) v_{B1}$$

$$v_{B2} = \left(\frac{m_B - m_A}{m_A + m_B} \right) v_{B1} + \left(\frac{2m_A}{m_A + m_B} \right) v_{A1}$$

Center of mass

(also valid for y and z)

$$x_{\text{cm}} = \frac{1}{M} \sum mx, \text{ where } M = \sum m, \text{ or}$$

$$= \frac{1}{M} \int x \, dm$$

$$v_{\text{cm},x} = \frac{1}{M} \sum mv_x$$

$$a_{\text{cm},x} = \frac{1}{M} \sum ma_x$$

Simple Harmonic Motion:

$$a_x = -\omega^2 x$$

$$\omega^2 = \frac{k}{m}$$

$$T = 2\pi \sqrt{\frac{m}{k}}$$

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

$$v_{\text{max}} = \sqrt{\frac{k}{m}} A$$

$$a_{\text{max}} = \frac{k}{m} A$$

Retarding force:

$$F = -bv$$

Rotation

$$\theta = \frac{s}{r}$$

$$\omega = \frac{d\theta}{dt}$$

$$\alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$$

$$v_{\text{tangential}} = r\omega$$

$$a_{\text{centripetal}} = r\omega^2 = \frac{v^2}{r}$$

$$a_{\text{tangential}} = r\alpha$$

Rotational kinematic equations:

$$\omega_f = \omega_i + \alpha t$$

$$\Delta\theta = \omega_i t + \frac{1}{2} \alpha t^2$$

$$\Delta\theta = \frac{(\omega_i + \omega_f)}{2} \times t$$

$$\omega_f^2 = \omega_i^2 + 2\alpha\Delta\theta$$

Moment of inertia:

$$I = \sum mr^2$$

$$I = \int r^2 dm$$

Parallel axis theorem

$$I = I_{\text{cm}} + Md^2$$

Rotational kinetic energy:

$$KE_{\text{rotational}} = \frac{1}{2} I \omega^2$$

Moment of inertia:

$$I = \sum m_i r_i^2$$

$$I = \int r^2 dm$$

Moment of inertia of hoop or thin cylindrical shell:

$$I_{\text{CM}} = MR^2$$

Moment of inertia of hollow cylinder:

$$I_{\text{CM}} = \frac{1}{2} M (R_1^2 + R_2^2)$$

Moment of inertia of solid cylinder:

$$I_{\text{CM}} = \frac{1}{2} MR^2$$

Moment of inertia of rectangular plate:

$$I_{CM} = \frac{1}{12} M(a^2 + b^2)$$

Moment of inertia of thin rod about its CM:

$$I_{CM} = \frac{1}{12} ML^2$$

Moment of inertia of thin rod about its end:

$$I = \frac{ML^2}{3}$$

Moment of inertia of solid sphere:

$$I_{CM} = \frac{2}{5} MR^2$$

Moment of inertia of spherical shell:

$$I_{CM} = \frac{2}{3} MR^2$$

Torque and Moment of force:

$$\tau = F \times r \times \sin \theta$$

$$\sum \tau = I\alpha$$

Work due to rotation:

$$\Delta W = \tau \times \Delta \theta$$

$$W = \int_{\theta_1}^{\theta_2} \tau d\theta$$

Power due to rotation:

$$P = \tau \times \omega$$

Pure rolling:

$$v_{cm} = r\omega$$

$$a_{cm} = r\alpha$$

$$KE_{\text{total}} = \frac{1}{2} Mv_{\text{cm}}^2 + \frac{1}{2} I_{\text{cm}} \omega^2$$

Conservation of Angular momentum:

$$\underbrace{\sum I\omega}_{\text{Before collision}} = \underbrace{\sum I\omega}_{\text{After collision}}$$

Avogadro's number:

$$N_A \approx 6.022 \times 10^{23} \text{ particles/mole}$$

Ideal gas constant:

$$R \approx 8.314 \text{ J mole}^{-1} \text{ K}^{-1}$$

Boltzmann constant:

$$k = \frac{R}{N_A} \approx 1.381 \times 10^{-23} \text{ J/K}$$

Ideal gas equation:

$$pV = nRT$$

Speed of a molecule

$$v_{\text{rms}} = \sqrt{\frac{3kT}{m}}$$

Mole

$$n = \frac{N}{N_A}$$

$$n = \frac{m}{M}$$

Absolute temperature

$$T = T_{\text{Celsius}} + 273.15$$

Specific heat c and molar heat C

$$Q = Mc(T_2 - T_1)$$

$$= nC(T_2 - T_1)$$

Latent heat L

$$Q = L \times M$$

First law of thermodynamics

$$\Delta U = Q - W$$

Internal energy for N molecules

$$\text{monatomic: } U = N \times \frac{3}{2} kT$$

$$\text{diatomic: } U = N \times \frac{5}{2} kT$$

Relationship between C_p and C_v for an ideal Gas

$$C_p - C_v = R$$

Isochoric process

$$\text{monatomic: } C_v = \frac{3}{2} R$$

$$\text{diatomic: } C_v = \frac{5}{2} R$$

C_p is undefined for isochoric process.

Isobaric process

$$\text{monatomic: } C_p = \frac{5}{2} R$$

$$\text{diatomic: } C_p = \frac{7}{2} R$$

C_v is undefined for isobaric process.

Isothermal process

$$W = nRT \ln \left(\frac{V_f}{V_i} \right)$$

Adiabatic process

$$\gamma = \frac{C_p}{C_v}$$

$$\text{monatomic: } \gamma = \frac{5}{3}$$

$$\text{diatomic: } \gamma = \frac{7}{5}$$

$$P_1 V_1^\gamma = P_2 V_2^\gamma$$

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$p_1^{1-\gamma} T_1^\gamma = p_2^{1-\gamma} T_2^\gamma$$

Thermal efficiency of a heat engine

$$e = 1 - \frac{|Q_C|}{|Q_H|}$$

Coefficient of performance of a heat pump for cooling

$$\text{COP} = \frac{|Q_C|}{|Q_H| - |Q_C|}$$

Entropy

$$\Delta S = \int_{\text{initial state}}^{\text{final state}} \frac{dQ}{T} \geq 0$$

Coulomb's Law:

$$F = \frac{k q_1 q_2}{r^2}$$

where

$$k = 8.9875 \times 10^9 \text{ Nm}^2/\text{C}^2$$

Electric field:

$$E = \frac{F}{q_0}$$

Electric field for point charge:

$$E = \frac{kq}{r^2}$$

Electric field at a distance along central axis of charged ring:

$$E = \frac{kQx}{(x^2 + a^2)^{3/2}}$$

Electric potential:

$$V = \frac{U}{q_0}$$

Electric potential in uniform field:

$$\Delta V = -Ed$$

Electric potential for point charge:

$$V = \frac{kq}{r}$$

Electric potential at a distance along central axis of charged ring:

$$V = \frac{kQ}{\sqrt{x^2 + a^2}}$$

Electric potential energy (for multiple point charges):

$$U = k \left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}} \right)$$

Electric current:

$$I = \frac{dQ}{dt}$$

$$I_{av} = nq v_d A$$

Ohm's Law:

$$I = \frac{\Delta V}{R}$$

Resistance of ohmic conductor:

$$R = \rho \frac{L}{A}$$

Resistivity:

$$\rho = \rho_0 [1 + \alpha(T - T_0)]$$

Electric power:

$$P = I\Delta V = I^2 R = \frac{\Delta V^2}{R}$$

Electric energy:

$$U = P\Delta t$$

Terminal voltage:

$$\Delta V = \mathcal{E} - Ir$$

Resistors in series:

$$R_{eq} = R_1 + R_2 + R_3 + \dots$$

Resistors in parallel:

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$

Kirchhoff's Junction Rule:

$$\Sigma I_{in} = \Sigma I_{out}$$

Kirchhoff's Loop Rule:

$$\Sigma \Delta V = 0$$

Magnetic force on a moving charge:

$$F_B = qvB \sin \theta$$

Magnetic force on a current carrying conductor:

$$F = ILB \sin \theta$$

Radius of moving charged particle in uniform magnetic field:

$$r = \frac{mv}{qB}$$

Angular speed:

$$\omega = \frac{qB}{m}$$

Period:

$$T = \frac{2\pi m}{qB}$$

Velocity selector:

$$v = \frac{E}{B}$$

Magnetic field due to long straight conductor:

$$B = \frac{\mu_0 I}{2\pi a} \text{ where}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ Tm/A}$$

Magnetic field at center of circular wire loop:

$$B = \frac{\mu_0 I}{2R}$$

Magnetic field at a distance along central axis of current-carrying ring:

$$B = \frac{\mu_0 I R^2}{2(x^2 + R^2)^{3/2}}$$

Magnetic force between parallel conductors:

$$F = \frac{\mu_0 I_1 I_2 l}{2\pi a}$$

Magnetic field inside toroid:

$$B = \frac{\mu_0 N I}{2\pi r}$$

Magnetic field inside solenoid:

$$B = \mu_0 n I$$

Magnetic flux:

$$\Phi_B = B A \cos \theta$$

Faraday's Law:

$$\mathcal{E} = -N \frac{d\Phi_B}{dt}$$

Motional emf:

$$\mathcal{E} = \frac{B l v}{R}$$

Power input:

$$P = F_{app} v = I l B v$$

Mass

Electron: 9.1094×10^{-31} kg

Proton: 1.6726×10^{-27} kg

Neutron: 1.6749×10^{-27} kg

Charge

Electron: -1.6022×10^{-19} C

Proton: $+1.6022 \times 10^{-19}$ C

Neutron: 0 C